

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

Date:	4 August 2026	Project Name:	Online Complaint Registration System (MERN Stack)	Max Marks:	4 Marks
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DATA FLOW DIAGRAMS (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD depicts system requirements graphically by illustrating how data enters and leaves the system, what processes transform the information, and where data is stored within the Online Complaint Registration platform.

Example: DFD Level 0 (Industry Standard) – Online Complaint Registration System

System Workflow Narrative

- **Complaint Submission:** Citizen inputs category, description, and location to log a new complaint into D1: Complaints database.
- **Assignment & Routing:** New complaint is validated and routed to the correct D2: Departments queue based on category; assigned to an available D3: Officers.
- **Resolution & Update:** Officer updates complaint status and adds resolution notes; changes sync back to the complaint record.
- **Confirmation & Notification:** Status change triggers the Notification Service, generating alerts for Citizen, Officer, and the Admin dashboard.

Citizen (User) → Officer → Notification Service → Admin / Dashboard
↓ Complaint Req / Status ↓ Assignment Alert ↓ Status Update Msg ↓ Report / Audit

1.0 File & Track Complaint 2.0 Process & Resolve Complaint

↔ Synchronizes with Data Stores ↔

D1: Complaints D2: Departments D3: Officers

USER STORIES TEMPLATE

Use the matrix below to capture all system capabilities, user requirements, and development priorities for the Online Complaint Registration system.

User Type	Functional Requirement	ID	User Story / Task	Acceptance Criteria	Priority	Release
Citizen (Mobile User)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account/ dashboard	High	Sprint-1
Citizen (Mobile User)	Registration	USN-2	As a user, I will receive a confirmation email once I have registered for the application.	I can receive confirmation email & click confirm	High	Sprint-1
Citizen (Mobile User)	Registration	USN-3	As a user, I can register for the application through Facebook.	I can register & access dashboard with Facebook	Low	Sprint-2
Citizen (Mobile User)	Registration	USN-4	As a user, I can register for the application through Gmail.	I can register & access dashboard with Gmail	Medium	Sprint-1
Citizen (Mobile User)	Login	USN-5	As a user, I can log into the application by entering email & password.	I can log into user dashboard successfully	High	Sprint-1
Citizen (Web User)	Complaint Filing	USN-6	As a citizen, I can file a complaint with a category, description, and location.	Complaint is logged with a unique tracking ID	High	Sprint-1
Citizen (Web User)	Complaint Tracking	USN-7	As a citizen, I can track my complaint status and view its timeline.	Real-time status update is shown on the dashboard	High	Sprint-2
Officer	Complaint Management	USN-8	As an officer, I can manage my assigned complaints and update their status.	Updated status reflects immediately for the citizen	High	Sprint-1
Support Staff	Escalation & Cancellation	USN-9	As support staff, I can assist citizens with cancelling or reopening complaints.	Cancellation/reopen logs are generated	Medium	Sprint-2
Administrator	Officer/Department Verification	USN-10	As an admin, I can review and assign complaints to verified officers/departments.	Only verified officers receive complaint assignments	High	Sprint-1